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Copper-catalyzed aerobic oxidation of aldehydes to carboxylic acids†

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An efficient, practical aerobic oxidation of aldehydes to carboxylic acids using O₂ or air as the oxidant has been developed. The reaction was carried out with a catalytic amount each of Cu(NO₃)₂·3H₂O and TEMPO at room temperature, tolerating many useful functional groups such as alkynyl, amino, aryl, etc. Complex molecules and compounds containing multiple aldehyde groups are applicable to this reaction with excellent selectivity. Gram-scale reactions have been easily conducted.

Introduction

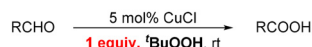
Carboxylic acids are a class of compounds that are extremely common in nature and widely used in laboratory,¹ daily life,² and industry.³ The oxidation of the corresponding aldehydes is one of the most straightforward strategies for the production of carboxylic acids.⁴ Traditionally, stoichiometric oxidants, such as metallic oxidants,⁵ peroxides,⁶ and hypervalent iodine compounds,⁷ are required. However, there exists an issue of generating equal amounts of waste from the applied oxidants. In contrast, molecular oxygen is a clean and atom-economic oxidant and water is usually the only by-product. Therefore, using environmentally friendly and inexpensive molecular oxygen as an oxidant has a significant advantage.⁸ Such aerobic oxidations have been achieved with noble metal catalysts, usually requiring higher reaction temperatures or stoichiometric amounts of additives.⁹ As a comparison, Earth-abundant metal catalysts, such as Fe, Co, and Ni, can also catalyze this process.¹⁰ Copper, as an Earth-abundant metal, has been widely used in catalytic aerobic oxidation reactions; however, most oxidation reactions of alcohols stopped at the stage of the corresponding aldehydes,^{11,12} and there are relatively few reports on the oxidation of aldehydes to carboxylic acids.¹³ In 2008, Sekar's group reported a CuCl-catalyzed oxidation of aldehydes to carboxylic acids at room temperature; however, the reaction requires one equivalent of explosive

chemical, TBHP, as the oxidant (Scheme 1a).^{6d} In 2016, Li's group reported the Cu/NHC-catalyzed aerobic oxidation of aldehydes with one equivalent of NaOH, and the reaction needs to be carried out at 50 °C.^{13a} Recently, Wu's group reported a (CAAC)CuCl-catalyzed aerobic oxidation of aldehydes at 25 °C, in which the substrates are mainly aromatic aldehydes (Scheme 1b).^{13b} Herein, we wish to report a practical Cu(NO₃)₂·3H₂O and TEMPO catalyzed aerobic oxidation of aldehydes to carboxylic acids at room temperature (Scheme 1c).

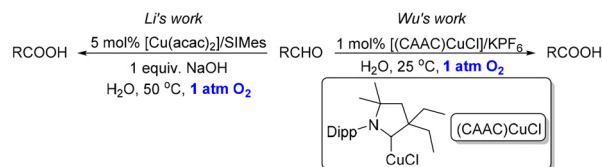
Results and discussion

We conducted initial studies with the evaluation of different copper catalysts to enable the aerobic oxidation of aldehydes

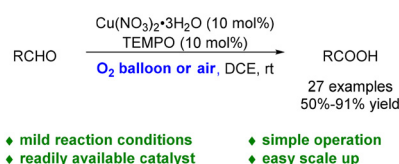
(a) Copper catalyzed oxidation of aldehydes to acids with TBHP as the oxidant



(b) Copper catalyzed aerobic oxidation of aldehydes to acids with the preprepared catalyst



(c) This work: Copper catalyzed aerobic oxidation with a commercially available catalyst



Scheme 1 Copper catalyzed oxidation of aldehydes to acids.

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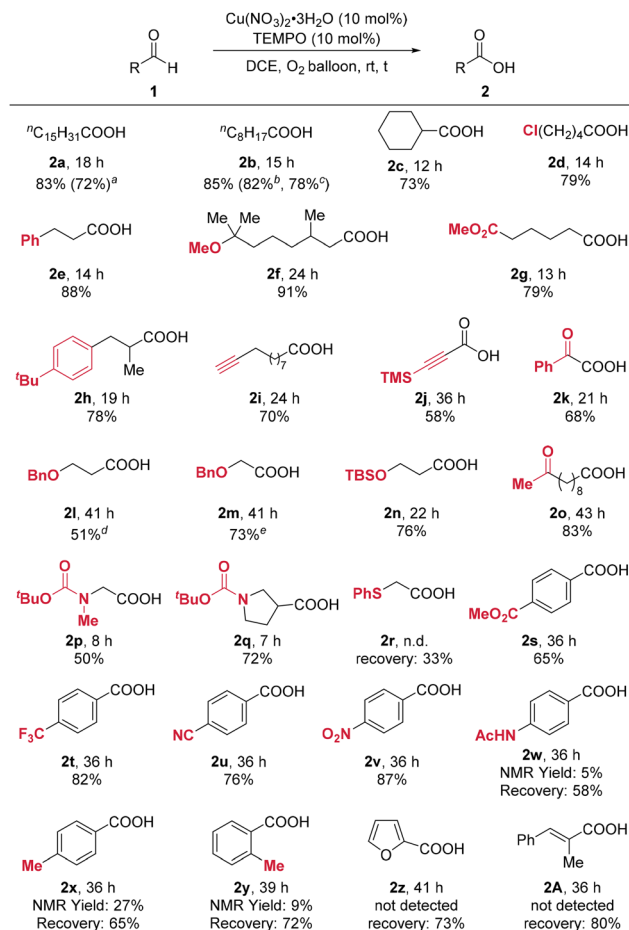
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to carboxylic acids using *n*-hexadecanal (**1a**) as the model substrate (Table 1). The oxidation of *n*-hexadecanal proceeded smoothly when using 10 mol% each of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ and TEMPO as the catalysts (Table 1, entry 1). When switching to other copper catalysts, such as CuCl_2 , $\text{Cu}(\text{OAc})_2$, CuBr , and CuOAc , the oxidative reaction did not occur (Table 1, entries 2–5). The reaction with LiNO_3 , NaNO_3 , or KNO_3 in the absence of any Cu salt failed to afford **2a**; thus, Cu must have played a critical role in the oxidation process (Table 1, entries 6–8). Combining LiNO_3 with other $\text{Cu}(\text{II})$ salts, such as CuCl_2 and $\text{Cu}(\text{OAc})_2$, did not yield the corresponding product (Table 1, entries 9 and 10). The reaction also failed with the more cost-effective 4-OH-TEMPO (Table 1, entry 11). No desired product was observed in toluene or MeCN (Table 1, entries 12 and 13). Finally, control experiments showed that the oxidative reaction proceeded very slowly without TEMPO (Table 1, entry 14), while it failed to proceed in the absence of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (Table 1, entry 15).

With the optimized reaction conditions in hand, the substrate scope was then investigated (Scheme 2). Linear aliphatic aldehydes and cyclohexanecarbaldehyde could all be oxidized smoothly to the corresponding acids **2a–2c**. The oxidation reaction of **1a** with an air balloon gave product **2a** albeit with a slightly lower yield. The reaction may be easily conducted on a 50 mmol scale by using a pure oxygen bag with **1b** as the substrate. When **1b** was reacted under an air atmosphere, similar results could be obtained by extending the reaction time. The reactions of aliphatic aldehydes containing different functional groups, such as halogen, ester, ether, and aryl groups,



Scheme 2 Substrate scope. The reaction was conducted with **1** (1.0 mmol), $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (0.1 mmol), and TEMPO (0.1 mmol) in 4 mL of DCE at rt for a specified time with an O_2 balloon. ^aAn air balloon was used. ^b**1b** (50 mmol) with an O_2 bag, 24 h. ^c**1b** (50 mmol) under open air, 30 h. ^d15% NMR yield of benzaldehyde and 22% recovery of **1l** were detected. ^e5% NMR yield of benzaldehyde was detected.

Table 1 Optimization of the reaction conditions^a

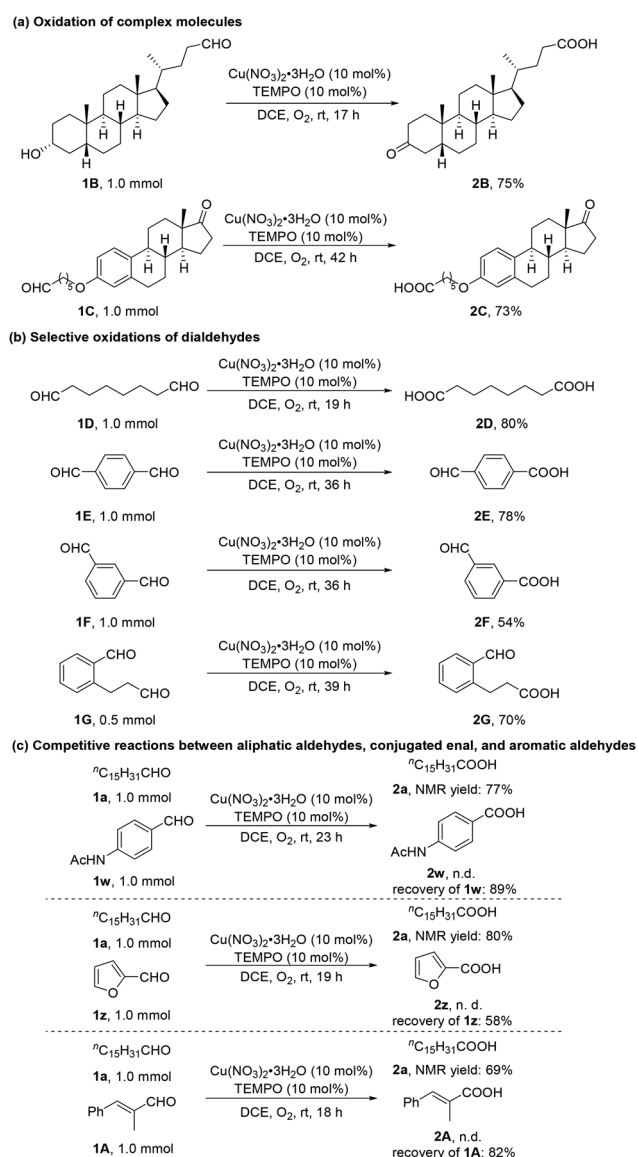
$^{13}\text{C}_{15}\text{H}_{31}\text{CHO}$ 1a , 0.5 mmol		$\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (10 mol%) TEMPO (10 mol%) DCE, O_2 balloon, rt, 18 h "standard conditions"	$^{13}\text{C}_{15}\text{H}_{31}\text{COOH}$ 2a
Entry	Variation from the "standard conditions"	Recovery (%)	Yield (%)
1	None	0	86
2	CuCl_2 instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	91	0
3	$\text{Cu}(\text{OAc})_2$ instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	89	0
4	CuBr instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	88	0
5	CuOAc instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	85	0
6	LiNO_3 instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	91	0
7	NaNO_3 instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	86	0
8	KNO_3 instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	93	0
9	CuCl_2 and LiNO_3 instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	88	0
10	$\text{Cu}(\text{OAc})_2$ and LiNO_3 instead of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	88	0
11	4-OH-TEMPO instead of TEMPO	84	0
12	Toluene instead of DCE	88	0
13	MeCN instead of DCE	88	0
14	Without TEMPO	31	38
15	Without $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	83	0

^aThe reaction was conducted with aldehyde **1a** (0.5 mmol), copper salts and/or nitrate salts (0.05 mmol), TEMPO (0.05 mmol), and solvent (2.0 mL) with an O_2 balloon at rt. The NMR yield and recovery were determined by ^1H NMR analysis with dibromomethane as the internal standard.

also afforded the corresponding carboxylic acids **2d–2h**. No benzylic C–H bond oxidation products were observed under the standard conditions (**2e** and **2h**). The reaction also tolerated conjugated or unconjugated C–C triple bonds (**1j** and **1i**) or the ketone functionality (**1k** and **1o**) affording carboxylic acids **2i–2k** and **2o**. The aerobic oxidation of 3-(benzyloxy)propanal **1l** afforded 51% of carboxylic acid **2l** and 15% of benzaldehyde with 22% recovery of **1l**. Such a benzylic C–O bond oxidative cleavage product was also formed in our previous report.¹⁴ The aerobic oxidation of 2-(benzyloxy)acetaldehyde **1m** afforded 73% of **2m** and also 5% of benzaldehyde. The aldehyde with silyl ether motif **1n** could also be converted to the corresponding acid **2n** in 76% yield; the corresponding deprotection product and the deprotective oxidation product were not obtained.¹⁵ Amino-carbonyl group-substituted aldehydes **1p** and **1q** also worked smoothly yielding the corresponding carboxylic acids **2p** and **2q**. However, for the substrate containing thioether unit **1r**, the reaction was messy without affording the desired acid **2r**. Aromatic aldehydes **1s–**

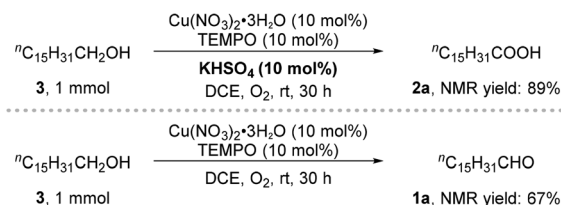
1v substituted with electron-withdrawing groups were more easily oxidized to afford the corresponding acids **2s–2v** in good to high yields. However, aldehydes **1w–1y** substituted with electron-donating groups afforded 5% of acid **2w** with 58% recovery, 27% of acid **2x** with 65% recovery, and 9% of acid **2y** with 72% recovery, respectively. The reactivity of electron-deficient aromatic aldehydes was much higher (**2s–2v** vs. **2w–2y**). The reactions of 2-furaldehyde **1z** and α,β -unsaturated enal **1A** also failed to afford the corresponding acids.

This reaction also works with complex molecules: compound **1B** containing a steroid skeleton could be oxidized to **2B** in 75% yield with the secondary alcohol being converted to the ketone as well; compound **1C** containing an estrone skeleton could be oxidized to the corresponding carboxylic acid **2C** in 73% yield (Scheme 3a). No epimerization of the stereocenters occurred owing to the mild reaction conditions, as deter-

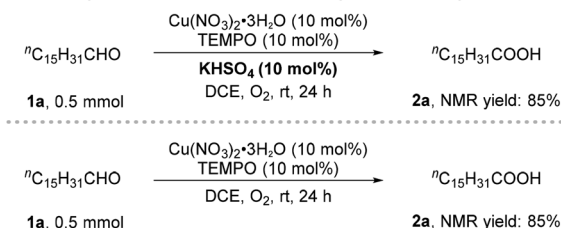


Scheme 3 Reactions of complex molecules, dialdehydes and competitive experiments.

(a) Copper-catalyzed aerobic oxidation of primary alcohols to carboxylic acids



(b) Copper catalyzed aerobic oxidation of aldehydes to carboxylic acids



Scheme 4 The effects of KHSO_4 on Cu/TEMPO catalyzed aerobic oxidation.

mined by ^1H and ^{13}C NMR analysis. Octanedial **1D** may be oxidized to suberic acid **2D**. Interestingly, for the reactions of *para*-phthalaldehyde **1E** and *meta*-phthalaldehyde **1F**, only one aldehyde group was oxidized to afford aromatic carboxylic acids **2E** and **2F** with another aldehyde functionality remaining. The observed phenomenon could be rationalized by the low solubility of **2E** and **2F**, leading to precipitation during the reaction. With the substrate containing both aromatic aldehyde and aliphatic aldehyde, the aliphatic aldehyde unit was preferentially oxidized to afford product **2G** (Scheme 3b). The result was consistent with the oxidation of *o*-methylbenzaldehyde **1y** (Scheme 2); the reactivity of *ortho*-alkyl substituted benzaldehyde was low due to the steric and electronic effects. Intermolecular competition experiments showed that aliphatic aldehydes were preferentially oxidized over aromatic aldehydes (Scheme 3c). When aliphatic aldehyde **1a** and aromatic aldehydes **1w** or **1z** were introduced to the same reaction vessel, only aliphatic aldehyde **1a** was oxidized; for the reaction of **1a** in the presence of α,β -unsaturated enal **1A**, only **1a** was oxidized as well.

In the copper-catalyzed primary alcohol oxidation to acids, the addition of KHSO_4 significantly accelerated the reaction.¹⁶ In the absence of KHSO_4 , only the generation of aldehyde was observed (Scheme 4a). However, in this copper-catalyzed aldehyde oxidation to acids, after 12 hours, regardless of the presence or absence of KHSO_4 , aldehyde **1a** was completely converted to carboxylic acid **2a** (Scheme 4b). Based on these results, we reasoned that the mechanism is most probably different from that of alcohol. Thus, the mechanism is not clear yet, and further studies are in progress.

Conclusions

In summary, we have developed an efficient synthesis of carboxylic acids *via* aerobic oxidation of aldehydes using

10 mol% each of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ and TEMPO as catalysts and ambient 1 atm of pure O_2 or oxygen in air as the oxidant. The transformation proceeded under very mild reaction conditions, which enabled the tolerance of many functionalities. This method exhibits excellent selectivity for different aldehyde functionalities. Further studies are being pursued in our laboratory.

Author contributions

S. M., H. Q. and Y. Y. conceived and directed the project and designed the experiments. Y. H. conducted most of the experiments. S. J. conducted the experiments for the paper revisions according to the review reports. Z. Z. participated in the optimization of the reaction conditions. Y. H., H. Q. and S. M. checked the experimental data and wrote the manuscript.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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